



Ideas for India: Innovation Challenge 2026 Common Application form

Instructions:

- Each response field in the form has a specific word limit, and supporting files must not exceed 25 MB per file.
- The number of questions in the presentation should be limited to the outline provided – do not add any additional questions. If additional questions are added, then the same will be deleted to stick to the format and Management will not be held responsible in any manner
- The font type and size used to fill the presentation should be Calibri - Size 12

Section 1: Applicant details

Applicant or Team Member Name(s):	Gaurav Barhate (individual applicant)
Institution/College/Company/Startup Name:	Indian Institute of Information Technology (IIIT), Vadodara
Project / Solution Title:	CarbonSathi — an AI-powered carbon accounting and CCTS/CBAM compliance assistant for India's MSMEs
Target Audience*	Indian MSMEs in manufacturing and export supply chains; suppliers of CCTS-obligated companies; MSME cluster associations; sustainability consultants and carbon verifiers
Key Markets / Regions Served (e.g., Cities / States / Countries): (<i>Applicable only for Startups and Enterprises</i>)	India. Pilot planned in the Morbi ceramic cluster (Gujarat) and the Tiruppur textile cluster (Tamil Nadu), then wider rollout. Pre-startup stage.
Estimated Implementation budget:	₹12 lakh for a 12-month pilot: cloud and AI infrastructure ₹2.5L, vernacular localization ₹1.5L, field onboarding and training in two clusters ₹4L, travel ₹1.5L, methodology and verification advisory ₹1.5L, contingency ₹1L
Select Type (highlight as applicable) *	☐ Student (recent graduate, Class of 2026) ☐ Professor ☐ Academician ☐ Professional ☐ Entrepreneur ☐ Other (please specify): _____
Problem Statement Applied Under (Select One)*	Sustainable & Resilient India Developing climate-resilient infrastructure, clean energy solutions, and circular economy models for long-term sustainability.
Select the Industry your Solution Primarily Targets: (highlight as applicable) *	Sustainable & Resilient India • Climate Tech & Carbon Management • Sustainable Infrastructure & Smart Cities • Electric Mobility & Clean Transportation • Energy Storage



Ideas for India: Innovation Challenge 2026 Common Application form

	- Green Hydrogen - Carbon Capture Climate mitigation and Adaptation - Nature-Based Solutions - Sustainable Agriculture
Annual Revenue (INR Crores) <i>(Applicable only if you have selected "Entrepreneur" under select type)</i>	☐ 2024 – 2025: _____ ☐ 2025 – 2026: _____

☒ I confirm that all applicant(s) are Indian nationals above 18 years of age (as on May 31, 2026).

Section: 2 – Case study details

Q.1. Describe the problem or challenge your solution seeks to address. Why is this problem important for India, and who is affected by it? (300 words limit).
India has started pricing carbon, and the clock is already running. From FY 2025-26, around 490 large industrial plants across seven sectors carry legally binding emission-intensity targets under the Carbon Credit Trading Scheme. Their first compliance filing is due on 31 July 2026, and the penalty for a shortfall is twice the market price of a carbon credit. On the export side, Europe's CBAM moved into its paid phase on 1 January 2026. Roughly €8–9 billion of Indian exports now carry a carbon cost, and GTRI expects coal-based steel and aluminium to lose 15–22% in effective price terms. Big companies will manage. The pressure lands on their suppliers, who cannot. India has 8.7 crore Udyam-registered MSMEs. Together they produce 31.1% of GDP, account for 48.6% of exports, and employ 38.9 crore people. Almost none of them can tell you their carbon footprint. A consultant engagement costs more than most small units spend on all their paperwork in a year. The software large companies use is priced for large companies, takes months to set up, and speaks only English. Meanwhile, a tile maker in Morbi keeps his records as paper bills and photos on a phone. He has no sustainability team. He has himself. So two things go wrong at once. Export-linked units start losing orders because they cannot produce the emissions data their buyers now demand. And the country cannot reach its NDC promise of cutting emission intensity by 45% by 2030 while a third of the economy stays unmeasured. The people affected are millions of small factory owners and the 38.9 crore workers who depend on them. You cannot manage what you cannot measure, and today most of India's industrial base cannot measure.
Q.2. Describe your idea (e.g., design thinking approach, community-led model, product or process innovation, etc). Additionally, how is your idea different, more practical, or more inclusive than existing solution and how does your solution support India's sustainability, resilience or net-zero ambitions? (Max 300 words)



Ideas for India: Innovation Challenge 2026 Common Application form

CarbonSathi is a carbon accounting and compliance assistant for small factories. The owner photographs an electricity bill or a fuel invoice, in whatever language it comes in, and the platform does the rest.

Behind that simple front are four steps. AI reads the uploaded document and pulls out the consumption data. An India-specific calculation engine applies the right factors, including the current CEA grid factor of 0.7117 tCO₂/MWh plus IPCC and India GHG Program values, to compute Scope 1, 2 and 3 emissions. The owner gets a live dashboard and reports shaped for CCTS-style compliance and for buyers asking CBAM questions. The system then suggests practical savings ranked by payback period, tracks water use and material waste, and shows how ready the unit is to earn carbon credits.

I designed it around how small units actually work. Records live on paper and WhatsApp. Nobody has time to learn carbon accounting, so the software absorbs that burden instead.

The existing options all miss this market. Persefoni and Watershed are built and priced for large corporates. Indian tools like sentra.world, Greenio and Zoei still assume a sustainability officer who works in English. Consultants do good work but cannot serve lakhs of units continuously. Nobody combines AI document reading, Indian languages, India-specific factors, compliance-ready output and a price of a few hundred rupees a month. That combination is the idea.

It is also the inclusive path to net zero. Measurement is the first step to reduction. If crores of small businesses can finally measure, India's carbon market reaches the base of the economy, where the NDC targets will actually be won.

Q.3. Explain how your solution contributes to the selected challenge theme. (Max 300 words)

If applying under Sustainable & Resilient India:

How does your solution contribute to climate resilience, clean energy adoption, resource efficiency, circular economy, or sustainable infrastructure?

Describe your solution and explain its environmental, social, and economic benefits, including its potential to improve resource efficiency, reduce environmental impact, enhance resilience to climate-related challenges, and support sustainable development outcomes at scale.

CarbonSathi builds the measurement, reporting and verification layer that India's smallest factories currently lack — the foundation of carbon management.

Clean energy: the recommendation engine finds electricity-heavy processes and shows the payback on rooftop solar and efficient motors. BEE has proven this approach, facilitating 800 kWp of rooftop solar across 18 ceramic plants in Thangadh, with typical paybacks under three and a half years. My platform turns that kind of one-off programme into a standing recommendation every unit sees.



Ideas for India: Innovation Challenge 2026 Common Application form

Resource efficiency: dashboards show energy, water and material use per unit of production. BEE's SME audits in Morbi and Tiruppur repeatedly found double-digit savings potential. An audit happens once; software watches every month.

Climate resilience: a unit that can produce emissions data keeps its buyers, avoids penalties and stays in the supply chain. That is resilience in its most practical form.

Sustainable infrastructure: over time I want the factor engine and methodology to become open rails others can build on, the way UPI and ONDC opened payments and commerce.

Environmental benefits: the first-year pilot targets about 35,000 tCO₂e measured and 2,500 tCO₂e cut annually across 200 units on conservative assumptions, alongside roughly 2.2 GWh of enabled solar generation and over 100 million litres of water saved in textile units.

Social benefits: the MSME sector employs 38.9 crore people, and protecting export orders protects those jobs. The pilot also trains local youth as "Carbon Mitras" and onboards women-led units first.

Economic benefits: 5–15% energy savings per unit, easier access to green finance, and future credit revenue once carbon trading opens to smaller players.

Scale: cluster by cluster, then state, then national, pulled along by CCTS growing toward more than 740 obligated entities whose suppliers will all need this data.

Q.4. Mention the current development stage of your solution. Also, is it tested or validated? (300 words limit)

- ☐ Idea Stage
- ☐ Proof of concept
- ☒ **Functional Prototype**
- ☐ MVP
- ☐ Simulation
- ☐ Pilot-Tested Solution
- ☐ Market-Ready

Briefly describe the same:

The platform runs today. It has a React frontend, Node.js APIs, MongoDB, and login secured with JWT and OAuth. The calculation engine carries the current CEA grid factor (0.7117 tCO₂/MWh, database version 21.0) along with IPCC and India GHG Program fuel factors. The AI module reads uploaded electricity bills and fuel invoices and turns them into structured consumption data. There is a live dashboard and there are PDF reports that mirror the data fields of a CCTS Form A.



Ideas for India: Innovation Challenge 2026 Common Application form

On validation: I cross-checked the engine's output against published CEA and BEE reference values, and tested the AI extraction on simulated datasets built around two typical units, a ceramic kiln and a textile dyeing house. The full flow works end to end, from upload to finished report.

What I have not done yet, and will not pretend otherwise: there is no live commercial pilot, no integration with accredited verifiers or the carbon market portal, and language coverage is still partial. The next step is a 90-day pilot with 15–20 units through a cluster association in Morbi or Tiruppur, turning simulated validation into field evidence.

Why believe I can execute: I have already shipped two production platforms on this exact stack — a real-time collaborative coding and interview product with AI evaluation built in, and a fault-tolerant webhook security service that survived concurrent load without losing a single payload. Real-time dashboards, AI pipelines and secure multi-tenant systems are what I have actually built, and they are precisely what CarbonSathi needs.

Q5. What is innovative about your solution?

Describe the key innovation(s) that make your solution unique. Innovation may be in the form of technology, product design, business model, process, delivery mechanism, partnerships, community engagement, or any other approach.

Please explain:

- What makes your solution different from existing alternatives? **Mention the existing alternatives used in comparison**
- How does it address the identified problem more effectively, efficiently, affordably, or inclusively?
- **What is the most novel aspect of your solution? Does your innovation reduce cost, improve efficiency or increase impact?**

What exists today, honestly compared: global platforms like Persefoni, Watershed and Sweep are excellent for corporates and priced accordingly, with long deployments and no CCTS awareness. Indian tools such as sentra.world, Greenio and Zoei ESG sit closer to our market but still assume an English-speaking sustainability person on staff. IIM Bangalore offers a free calculator, but it needs manual entry and produces no compliance outputs. Consultants are credible but expensive per engagement and cannot cover lakhs of units continuously. The real status quo is Excel, or nothing.

Why mine works where these do not: the biggest barrier is not awareness, it is data entry. A factory owner will not type bills into a form. So the AI reads the bills instead. Photograph, upload, done. On cost: a few hundred rupees a month against consultant engagements costing orders of magnitude more, roughly a hundred times cheaper per measured unit. On reach: a vernacular, mobile-first product gets to owners no English software will ever touch.



Ideas for India: Innovation Challenge 2026 Common Application form

The most novel piece is the pipeline itself: AI tuned to Indian documents, from state DISCOM bill formats to informal fuel invoices in regional languages, feeding straight into CCTS- and CBAM-shaped outputs, at a price a small unit can pay. Nobody occupies that quadrant today.

Does it cut cost, improve efficiency, increase impact? All three. Cost falls about 100× versus the consultant route. Measurement time drops from weeks to minutes. And a segment producing roughly 31% of GDP becomes visible to India's carbon system for the first time.

Q.6. Briefly mention the exact outcome of your solution that aims to achieve:

Metrics	2024-25	2023-24	Unit / % Improvement
No. of beneficiaries served (monthly/yearly)	-	-	Year-1 target: 200 MSME units across 2 clusters (~2,000 workers benefiting); Year-3: 50,000 units
Cost Savings / Efficiency Improvement (%)	-	-	5–15% energy savings per unit (basis: BEE SME cluster audits, Morbi & Tiruppur manuals); units adopting recommended rooftop solar cut electricity bills 50–70% (payback 2.8–3.4 years)
Jobs created	-	-	25–30 trained "Carbon Mitra" field roles in pilot year; 2,000+ by Year-3 through green-skilling of local and ITI youth
Environmental Impact	-	--	~35,000 tCO ₂ e/yr measured and ~2,500 tCO ₂ e/yr reduced across pilot units (basis: CEA grid factor 0.7117 tCO ₂ /MWh); ~2.2 GWh/yr renewable generation enabled; 100–150 million litres/yr water conserved in textile units; ~2,000

Ideas for India: Innovation Challenge 2026

Common Application form

			tonnes/yr ceramic waste diverted to reuse
Revenue / Economic Value Generated	-	-	SaaS at ₹500–2,000/month per unit; pilot revenue target ₹12–15 lakh; ₹4 crore+ combined annual energy savings unlocked for pilot units; future carbon-credit revenue once CCC trading opens (~Oct 2026, early estimates ₹250–1,500/tonne)
Other Relevant Metric	-	-	100% of pilot units CBAM/buyer-report ready; measurement time per unit cut from weeks (consultant audit) to minutes (AI upload); first-ever formal emissions baseline for every onboarded unit
Note - Please include only relevant metrics for your challenge			

Q.6.1. Additionally, explain the most significant impact created or expected in these categories. Use NA in case of no response (Max 150 words) – Environmental, social and economical aspects

Environmental: the pilot alone should bring about 35,000 tCO₂e under measurement and cut around 2,500 tCO₂e a year, alongside 2.2 GWh of enabled solar generation and over 100 million litres of water saved. At national scale this becomes millions of tonnes, feeding directly into the NDC's 45% intensity target.

Social: the MSME sector employs 38.9 crore people. Keeping small exporters inside carbon-conscious supply chains protects those livelihoods. The pilot trains local youth as Carbon Mitras and brings women-led units in first.

Economic: 5–15% off energy bills per unit, export orders preserved within ₹8–9 billion of CBAM-exposed trade, and a route to credit revenue once CCC trading begins, expected around October 2026 with early price estimates of ₹250–1,500 per tonne.

Q.7. Have you collaborated or partnered with any NGOs, government bodies, startups, corporates, or academic institutions, etc in supporting your solution? If yes, describe the nature of the collaboration and its contribution to the solution. (Max 300 words)



Ideas for India: Innovation Challenge 2026 Common Application form

Not yet, and I would rather say that plainly than dress it up. I am an individual applicant a few months out of college. What I do have is a specific plan for who I need and why.

Cluster associations come first: the Morbi ceramic manufacturers' association and the Tiruppur exporters' association, because they are the trusted door to hundreds of units. My first 30 days after this round are reserved for securing pilot letters of intent from them. District MSME facilitation offices come next, for official introductions and linkage to schemes like RAMP. BEE's SME programme matters for methodology: my factor engine follows their published cluster manuals so the numbers stand up to scrutiny. IIIT Vadodara, my alma mater, offers technical mentorship and student support for field work. Climate incubators such as Climate Collective are the route to funding and ecosystem access, and those applications are going out in parallel with this one.

Further ahead, accredited verifiers become integration partners, using the platform as their field-data layer.

If this challenge gives me one thing beyond validation, I hope it is the credibility to convert this pipeline into signed pilots faster.

Q.8. Explain the sustainability, scalability, replicability and Inclusivity plan of the solution and mention the steps taken to make it inclusive and replicable in the future* (300 words limit)

Money: freemium to start, then paid tiers between ₹500 and ₹2,000 a month for compliance reports and recommendations, then association-level licences where one contract covers hundreds of units, and eventually a small fee on the verification marketplace. Costs stay low because the product is cloud-native and AI does the work a consultant would.

Scale: the architecture is stateless and queue-based, patterns I have already run in production, so it scales horizontally without redesign. Distribution scales the same way: one association partnership onboards a whole cluster, keeping customer acquisition cost close to zero. Demand compounds on its own, because CCTS is heading toward more than 740 obligated entities whose suppliers will all be asked for emissions data.

Replication: the factor engine is configuration-driven. Adding a new sector or state means a new configuration, not a rebuild. I plan to publish the methodology openly so anyone can audit or adapt it. The long-term ambition is ONDC-style open rails for carbon data.

Inclusion, in concrete steps: an interface being built in 10+ Indian languages with voice guidance, mobile-first design that works on weak connections, onboarding that assumes zero prior knowledge, a hard price floor so the smallest units are never priced out, discounted tiers and first-priority onboarding for women-led MSMEs, and Carbon Mitra training that recruits local youth, including from ITIs.



Ideas for India: Innovation Challenge 2026 Common Application form

The impact sustains itself because the incentives line up. The platform earns from measurement while the savings go to the factory. The greener and more efficient a unit becomes, the more useful the product is to it. Growth and environmental outcome reinforce each other instead of trading off.

Q.9. Elevator pitch: Explain in a few lines why your idea deserves to be selected as a winner. Highlight what makes it unique, impactful, and scalable. (300 words limit)

Two dates changed the rules for Indian industry. On 1 January 2026, Europe started charging for the carbon in our exports. On 31 July 2026, India's own first carbon-compliance filings fall due. Between those dates, measuring carbon stopped being optional. Yet 8.7 crore MSMEs, producing 31% of our GDP, still cannot measure a single tonne.

My answer is simple: stop asking small factories to learn carbon accounting, and let AI do it. Photograph a bill in any Indian language and get Scope 1, 2 and 3 emissions, compliance-ready reports and ranked savings actions in minutes, for a few hundred rupees a month.

No global platform, Indian tool or consultant offers AI document reading, vernacular access, India-native factors and compliance output together at this price. The prototype already runs end to end, and I have shipped production AI platforms before this one.

Every measured MSME moves India closer to its 45% intensity cut and to net zero 2070. We cannot manage what six crore businesses cannot measure. CarbonSathi makes them count.

Section 3: Documents

Attach documents – Create a google drive link that is accessible to all and attached the same here

Documents to be attached in the google drive link:

- Concept-to-prototype execution plan deck
- Demonstration Evidence – if any (Attach a demo video link)
- Any other supporting

Attach google drive link here:

https://drive.google.com/file/d/1LWC2xdY_kiFmj-NDUXR6LMLbcYiqD1a3/view?usp=sharing

Declaration*

- ☒ I/We confirm that the idea is original and developed by the individual/team.
- ☒ I/We agree to abide by the terms and conditions of the solution Challenge.